

Scheme of Study BS Information Technology

#	Code	Pre-	Course Title	Dom	Related SDG	Cr Hr	Total	Semester
1	CSC-111		Programming Fundamentals	Core	4,9	3	3	1
2	CSC-111L		Programming Fundamentals	Core		1(0-3)	1	1
3	CS-101		Application of Information & Communication Technologies	GER	4	3	3	1
4	ENG-161		Functional English	GER	4	3 (3-0)	3	1
5	MTH-171		Calculus and Analytic Geometry	GER	4,8,9,16	3 (3-0)	3	1
6	PHY-151		Applied Physics	GER	4,9,12,15	2	2	1
7	PHY-151L		Applied Physics	GER		1(0-3)	1	1
8	QS-110		Teaching of Holy Quran with Translation-I/ Ethics-I	GER	4,10,16,17	1(1-0)	1	1
	MTH-101		Basic Math-I (Deficiency Course)	GER	4	*3		1
			Total CR				17	1
1	CSC-112	Programming Fundamentals	Object Oriented Programming	Core	1,2,4,9,16	3	3	2
2	CSC-112L	Programming Fundamentals	Object Oriented Programming	Core		1(0-3)	1	2
3	CSC-152	Applied Physics	Digital Logic Design	Core	4	2	2	2
4	CSC-152L	Applied Physics	Digital Logic Design	Core		1(0-3)	1	2
5	ENG-162	Functional English	Expository Writing	GER	4	3 (3-0)	3	2
6	CS-102		Discrete Structures	GER	4	3 (3-0)	3	2
7	MTH-172	Calculus and Analytic Geometry	Linear Algebra	Maths	4	3 (3-0)	3	2
	MTH-102		Basic Math-II (Deficiency Course)	GER	4	*3		2
			Total CR				16	2
1	CSC-213	Object Oriented Programming	Data Structures	Core	4,9	3	3	3
2	CSC-213L	Object Oriented Programming	Data Structures	Core		1(0-3)	1	3
3	CSC-253	Digital Logic Design	Computer Organization & Assembly Language	Core	4	2	2	3
4	CSC-253L	Digital Logic Design	Computer Organization & Assembly Language	Core		1(0-3)	1	3
5	MTH-273		Probability & Statistics	Maths	4,8,9,12,16	3 (3-0)	3	3
6	ENG-263	Expository Writing	Technical & Business Writing	EW	4	3 (3-0)	3	3
7	IST-201		Islamic Studies	GER	4,16	2 (2-0)	2	3
8	PST-201		Ideology and Constitution of Pakistan	GER	4,5,16,16,17	2 (2-0)	2	3
9	QS-210		Teaching of Holy Quran with Translation-II/ Ethics-II	GER	4,10,16,17	1(1-0)	1	3
			Total CR				18	3
1	CSC-221	Data Structures	Database Systems	Core	4,9	3	3	4

2	CSC-221L	Data Structures	Database Systems	Core		1(0-3)	1	4
3	CSC-215	Data Structures	Operating Systems	Core	4	2	2	4
4	CSC-215L	Data Structures	Operating Systems	Core		1(0-3)	1	4
5	CSC-214	Data Structures	Analysis of Algorithms	Core	4,9,11,12	3 (3-0)	3	4
6	MTH-274	Calculus and Analytic Geometry	Multivariable Calculus	Maths	4	3 (3-0)	3	4
7	SS-201		Financial Accounting	SS	1,8,9	3 (3-0)	3	4
	ITE-2XXX		Domain Elective-I	Domain Elective-I		3 (3-0)	3	
			Domain Elective-I					
8	ITE-2901		E-Commerce	Domain Elective-I	4,9	3 (3-0)		4
	ITE-2902		Multimedia Systems and Design	Domain Elective-I	4	3 (3-0)		4
			Total CR				19	4
1	CSC-331	Data Structures	Computer Networks	Core	4	2	2	5
1	CSC-331L	Data Structures	Computer Networks	Core		1(0-3)	1	5
2	CSC-381		Software Engineering	Core	3,4,7,8,9,11,12,13,16,17	3 (3-0)	3	5
3	ITE-39XX		Domain Elective-II	Domain Elective-II		3 (3-0)	3	5
4	ITC-322		DB Administration & Management	Domain Core	4,8,9,11,16	2	2	5
4	ITC-322L		DB Administration & Management	Domain Core		1(0-3)	1	5
5	CSC-315	Data Structures	Artificial Intelligence	Core	4,8,9,11,13,16	2	2	5
5	CSC-315L	Data Structures	Artificial Intelligence	Core		1(0-3)	1	5
	ITE-39XX		Domain Elective-III	Domain Elective-III		3 (3-0)	3	5
	QS-310		Teaching of Holy Quran with Translation-III/ Ethics-III	GER	4,10,16,17	1(1-0)	1	5
			Domain Elective-II & III					
	ITE-3903		Web Engineering	Domain Elective	4,8,9,10,17	3 (3-0)		5
6	ITE-3904		Software Project Management	Domain Elective	4,8,9,12	3 (2-3)		
	ITE-3905	Computer Networks	Network Security	Domain Elective	4,8,9	3 (3-0)		5
	ITE-3906	Software Engineering	Software Requirement Engineering	Domain Elective	4,9,11,12	3 (3-0)		5
			Total CR				19	5
1	CSC-332		Information Security	Core	4,8,9,16	2	2	6
1	CSC-332L		Information Security	Core		1(0-3)	1	6
2	ITE-39XX		Domain Elective-IV	Domain Elective-IV		3 (3-0)	3	6
3	ITC-333	Computer Networks	System & Network Administration	Domain Core	4,9	2	2	6
3	ITC-333L		System & Network Administration	Domain Core		1(0-3)	1	6
4	ITE-3XXX		Domain Elective-V	Domain Elective-V		3 (2-3)	3	6

5	ITE-3XXX		Domain Elective-VI	Domain Elective-VI		3 (3-0)	3	6
6	ITC-343	Web Engineering	Web Technologies	Domain Core	4,8,9,10,17	2	2	6
6	ITC-343L	Web Engineering	Web Technologies	Domain Core		1(0-3)	1	6
			Domain Elective-IV, V & VI					
	ITE-3308		HCI & Computer Graphics		4			
	ITE-3309		Visual Programming		4,8,9,10			
	ITE-3310		Cloud Computing		4,9			
	ITE-3311		Wireless Networks		4,9			
	ITE-3312		Digital Signal Processing		4,9			
	ITE-3313		Human Computer Interaction		3,4,8,9,10			
			Total CR				18	6
1	FYP-491		Final Year Project - I	Core		3 (3-0)	3	7
2	ITC-434		Cyber Security	Domain Core	16	2	2	7
2	ITC-434L		Cyber Security	Domain Core		1(0-3)	1	7
3	ITC-435		Information Technology Infrastructure	Domain Core	4,8,9,11,17	2	2	7
3	ITC-435L		Information Technology Infrastructure	Domain Core		1(0-3)	1	7
4	PP-401		Professional Practices	GER	4,8,10,16,17	2 (2-0)	2	7
5	ITE-4XXX		Domain Elective-VII	Domain Elective-VII		3 (2-3)	3	7
	QS-410		Teaching of Holy Quran with Translation-IV	GER	4,10,16,17	1(1-0)	1	7
			Domain Elective-VII					
	ITE-4114		Mobile Application Development		4,8,9,10,11,13			
	ITE-4015		Enterprise Systems		8,9,12			
	ITE-4016		Business Intelligence		8,9,12,17			
			Total CR				15	7
1	FYP-492	FYP-I	Final Year Project - II	Core		3 (3-0)	3	8
2	ITC-416	Artificial Intelligence	Introduction to Data Science	Core	4,9	2	2	8
2	ITC-416L	Artificial Intelligence	Introduction to Data Science	Core		1(0-3)	1	8
3	MGT-401		Social Sciences (Example: Introduction to Management)	GER	4,8,9	2 (2-0)	2	8
4	MGT-402		Entrepreneurship	GER	4,8,9	2 (2-0)	2	8
5	MGT-403		Civics and Community Engagement	GER	4,11	2 (2-0)	2	8
			Total CR				12	8

Total CR	134
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Generic Structure for Computing Disciplines:		
Areas	Credit Hours	Courses
Computing Core	46	14
Domain Core	18	6
Domain Elective	21	7
Mathematics & Supporting Courses	12	4
Elective Supporting Courses	3	1
General Education Requirement	34	16
Totals	134	48

Sr.	Course Code	Course Title	Dom		Cr Hr
1	ITE-2001	E-Commerce	Domain Elective		3 (2-3)
2	ITE-2002	Multimedia Systems and Design	Domain Elective		3 (2-3)
3	ITE-3403	Web Engineering	Domain Elective		3 (2-3)
4	ITE-3804	Software Project Management	Domain Elective		3 (2-3)
5	ITE-3305	Network Security	Domain Elective		3 (2-3)
6	ITE-3806	Software Requirement Engineering	Domain Elective		3 (2-3)
7	ITE-3107	Virtual Systems & Services	Domain Elective		3 (2-3)
8	ITE-3308	HCI & Computer Graphics	Domain Elective		3 (2-3)
9	ITE-3309	Visual Programming	Domain Elective		3 (2-3)
10	ITE-3310	Cloud Computing	Domain Elective		3 (2-3)
11	ITE-3311	Wireless Networks	Domain Elective		3 (2-3)
12	ITE-3312	Human Computer Interaction	Domain Elective		3 (2-3)
13	ITE-3313	Digital Signal Processing	Domain Elective		3 (2-3)
14	ITE-4114	Mobile Application Development	Domain Elective		3 (2-3)
15	ITE-4015	Enterprise Systems	Domain Elective		3 (2-3)
16	ITE-4016	Business Intelligence	Domain Elective		3 (2-3)

Course Category	Number
General	0

Programming Fundamentals	1
Data Base	2
Networking	3
Web	4
Architecture	5
English	6
Math	7
Software Development	8
Elective Course	9